Cmd 4

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- The California housing dataset is a widely used dataset in machine learning and is available in the scikit-learn library
- It contains information about housing prices in various districts of California. The dataset is often used for regression tasks to predict the median house value in a given district based on several features.
- **The California housing dataset provides the following information for each district:**

1. MedInc: Median income of households in the district.
2. HouseAge: Median age of houses in the district.
3. AveRooms: Average number of rooms per house.
4. AveBedrms: Average number of bedrooms per house.
5. Population: Total population in the district.
6. AveOccup: Average number of occupants per house.
7. Latitude: Latitude of the district's location.
8. Longitude: Longitude of the district's location.
9. MedHouseVal: Median value of houses in the district (the target variable).

- The goal of using this dataset is typically to build a regression model that can predict the median house value based on the given features.

Cmd 5

Select the search algorithm

The two main choices are:

- `hyperopt.tpe.suggest` : Tree of Parzen Estimators, a Bayesian approach that iteratively and adaptively selects new hyperparameter settings to explore based on previous results
- `hyperopt.rand.suggest` : Random search, a non-adaptive approach that samples over the search space

Run the tuning algorithm with Hyperopt fmin()

Cmd 27

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To distribute tuning, add one more argument to `fmin()` : Class `Trials` & Method `SparkTrials`

`SparkTrials` takes 2 optional arguments:

- `parallelism` : Number of models to fit and evaluate concurrently. The default is the number of available Spark task slots.
- `timeout` : Maximum time (in seconds) that `fmin()` can run. The default is no maximum time limit.

Cmd 28

Provide python code in data bricks environment for hyper optimization- use following steps

1. Load data set - import required packages (bumpy, fetch_california housing, Cross val score, SVC, RandomForestClassifier, LogisticRegression, hyper opt, mlfow),
2. Feature Engineering - scaling, numerical target column to discrete values
3. HyperOpt workflow - define function to minimize, search space over the hyper parameters, select search algorithm, run tuning algorithm
4. Print results of best parameters
5.